

Stokes Convection

Balance Equations:

$$1) \quad \nabla \cdot [\mu (\nabla \underline{v} + \nabla \underline{v}^T)] - \nabla p = \rho \underline{g}$$

$$2) \quad \nabla \cdot \underline{v} = 0$$

$$3) \quad \rho c_p \frac{\partial T}{\partial t} + \nabla \cdot [\underline{v} \rho c_p T - \kappa \nabla T] = 0$$

cool adiabatic
cooling faster

Constitutive equation: $\rho(T) = \rho_0 [1 - \alpha (T - T_0)]$

ρ_0 is reference density at T_0 $[\frac{M}{L^3}]$

$\alpha = \frac{1}{V} \left. \frac{\partial V}{\partial T} \right|_p$ coefficient of thermal expansion $[\frac{1}{K}]$

↑ generic Temp unit

Typical density change in ice:

$$\rho_0 = 920 \frac{kg}{m^3}$$

$$\alpha = 1.6 \cdot 10^{-4} \frac{1}{K}$$

$$\text{Europa: } \Delta T_E = 170 K \Rightarrow \Delta \rho_E = \rho_0 \alpha \Delta T_E \approx 25 \frac{kg}{m^3} \approx 3\%$$

$$\text{Glacier: } \Delta T_G = 40 K \Rightarrow \Delta \rho_G = \rho_0 \alpha \Delta T_G \approx 5 \frac{kg}{m^3} \approx 0.6\%$$

$\Rightarrow$ density change is small

Boussinesq Approximation:

Set $\rho = \rho_0$ everywhere except in ρg

→ buoyancy term

$$\rho_0 c_p \frac{\partial T}{\partial t} + \nabla \cdot [\underline{v} \rho_0 c_p T - \kappa \nabla T] = 0$$

Introduce thermal diffusivity: $k = \frac{\kappa}{\rho_0 c_p}$

$$\frac{\partial T}{\partial t} + \nabla \cdot [\underline{v} T - k \nabla T] = 0 \quad (\rho_0 c_p \text{ \& } \kappa = \text{const.})$$

Introduce reduced pressure: $\pi = p - \rho_0 g z$

$$\rho g = -\rho_0 g \nabla z \quad \hat{z} = \nabla z$$

$$-\nabla p + \rho_0 g \nabla z = -\rho_0 \alpha (T - T_0) \rho_0 g \nabla z$$

substituting:

$$\nabla \cdot [\mu (\nabla \underline{v} + \nabla^T \underline{v})] - \nabla \pi = \rho_0 \alpha (T - T_0) g \hat{z}$$

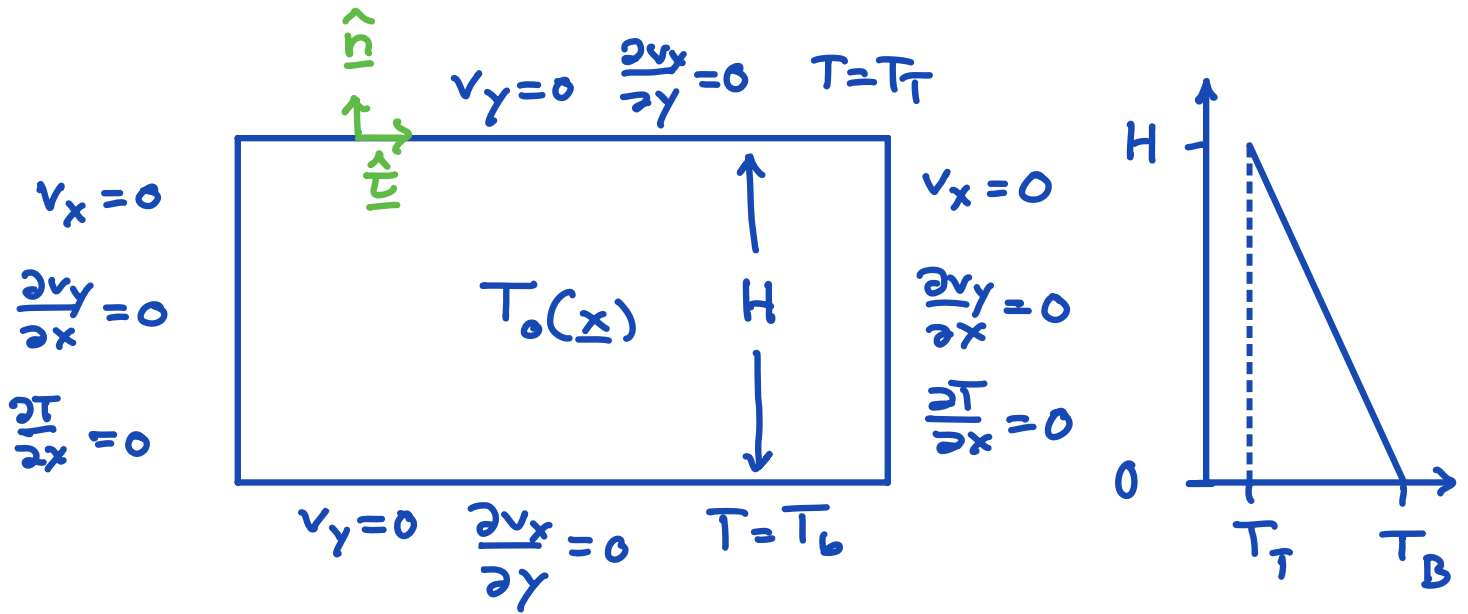
Governing Equations:

$$1) \quad \nabla \cdot [\mu (\nabla \underline{v} + \nabla^T \underline{v})] - \nabla \pi = \rho_0 \alpha (T - T_0) g \hat{z}$$

$$2) \quad \nabla \cdot \underline{v} = 0$$

$$3) \quad \frac{\partial T}{\partial t} + \nabla \cdot [\underline{v} T - k \nabla T] = 0$$

Initial & Boundary Conditions



$\hat{n}$ = out ward normal

$\hat{t}$ = tangent unit vector

Mass: no penetration

$$\underline{v} \cdot \underline{n} = 0$$

Momentum: free slip = no shear stress / traction

$$\begin{aligned} |\underline{t}_{||}| &= \hat{t} \cdot (\underline{\sigma} \hat{n}) = 0 \quad \text{every where} \\ &= \hat{t} \cdot (\nabla \underline{v}) \hat{n} = 0 \end{aligned}$$

Energy:

a) Prescribed temperature on top & bottom

$$T(y=0) = T_B \quad T(y=H) = T_T$$

b) no heat flow elsewhere: $\underline{q} \cdot \underline{n} = 0$

Scaling the equations

External scales: H, T_T, T_B, T_0

$$\underline{x}' = \frac{x}{H} \quad T' = \frac{T - T_T}{\underbrace{T_B - T_T}_{\Delta T > 0}} \quad \underline{v}' = \frac{v}{v_c} \quad t' = \frac{t}{t_c} \quad \pi' = \frac{\pi}{\pi_c}$$

$v_c, t_c, \pi_c \rightarrow$ internal scales

Many options $\rightarrow$ standard choice (Tackley p.5)

Energy equation:

$$\frac{\Delta T}{t_c} \frac{\partial T'}{\partial t'} + \frac{1}{H} \nabla' \cdot [v_c \Delta T \underline{v}' T' - \frac{k \Delta T}{H} \nabla' T'] = 0$$

$$\frac{\partial T'}{\partial t'} + \nabla' \cdot \left[\frac{v_c t_c}{H} \underline{v}' T' - \frac{k t_c}{H^2} \nabla' T' \right] = 0$$

choose $t_c = \frac{H^2}{k}$ diffusive timescale

$$\frac{\partial T'}{\partial t'} + \nabla' \cdot \left[\frac{v_c H}{k} \underline{v}' T' - \nabla' T' \right] = 0$$

choose $v_c = \frac{k}{H}$

$$\Rightarrow \frac{\partial T'}{\partial t} + \nabla \cdot [\underline{v}' T' - \nabla T'] = 0$$

Momentum balance:

$$\nabla \cdot [\mu (\nabla \underline{v} + \nabla \underline{v}^T)] - \nabla \pi = \rho_0 \alpha (T - T_0) g \hat{z}$$

$$\frac{\mu v_c}{H^2} \nabla \cdot [\nabla \underline{v}' + \nabla \underline{v}'^T] - \frac{\pi_c}{H} \nabla \pi' = \rho_0 \alpha [T_T - (T_B - T_T) T' - T_0] g \hat{z}$$

$$\nabla \cdot [\nabla \underline{v}' + \nabla \underline{v}'^T] - \frac{\pi_c H}{\mu v_c} \nabla \pi' = \frac{\rho_0 \alpha g H^2}{\mu v_c} [T_T - (T_B - T_T) T' - T_0]$$

choose $\pi_c = \frac{\mu v_c}{H} = \frac{\mu k}{H^2}$

$$\nabla \cdot [\nabla \underline{v}' + \nabla \underline{v}'^T] - \nabla \pi' = \frac{\rho_0 \alpha g H^3}{\mu k} [T_T - (T_B - T_T) T' - T_0]$$

Choose : $T_0 = T_T$

$$\text{rhs} = \frac{\rho_0 \alpha g H^3}{\mu k} [\cancel{T_T} - \overbrace{(T_B - T_T)}^{\Delta T} T' - \cancel{T_T}] = \frac{\rho_0 \alpha \Delta T g H^3}{\mu k} T' \hat{z}$$

Rayleigh number: $Ra = \frac{\rho_0 \alpha \Delta T g H^3}{\mu k} = \frac{\Delta \rho g H^3}{\mu k}$

$$\Delta \rho = \rho_0 \alpha \Delta T$$

$$\Rightarrow \nabla \cdot [\nabla \underline{v}' + \nabla \underline{v}'^T] - \nabla \pi' = Ra T' \hat{z}$$

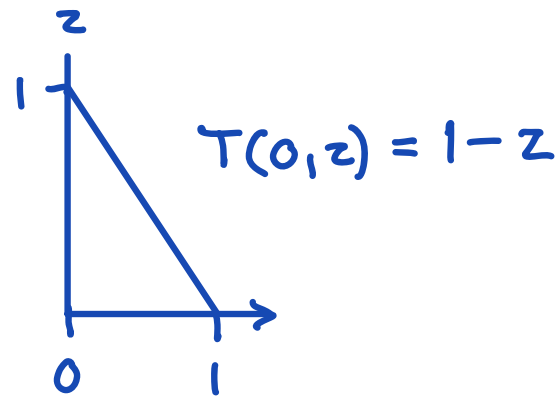
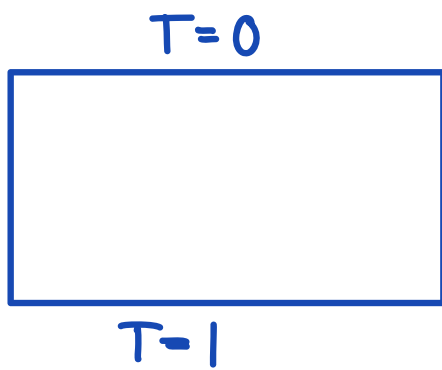
Dimensionless equations (dropped primes)

$$1) \nabla \cdot [\nabla \underline{v} + \nabla T] - \nabla \pi = Ra T \hat{z}$$

$$2) \nabla \cdot \underline{v} = 0$$

$$3) \frac{\partial T}{\partial t} + \nabla \cdot [\underline{v} T - \nabla T] = 0$$

Domain, BC's & IC



$\Rightarrow$ all other BC are homog. Neumann!

$\Rightarrow$ Single parameter: $Ra = \frac{\rho g H^3}{\mu k}$

Questions:

1) At what Ra does convection begin?

2) How does convective flux scale with Ra ?